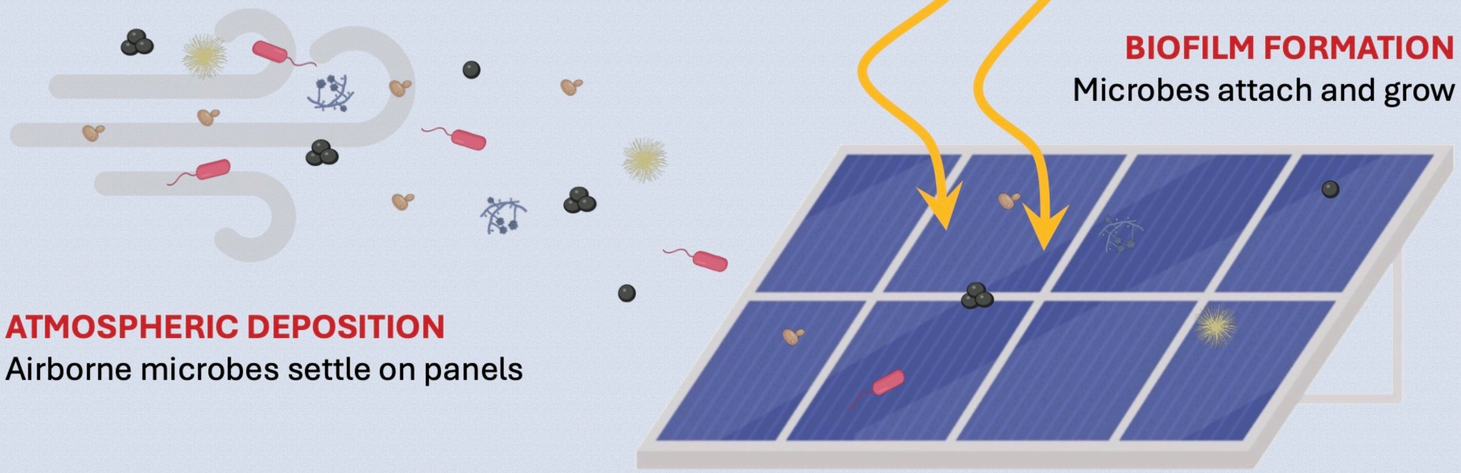
PreMiEr
Precision Microbiome Engineering
NSF Engineering Research Center

Solar Unsoiled

Erica Babusci, Duke University

PROBLEM

Biosoiling: Fungi colonize solar panels, and **less electricity** is produced.



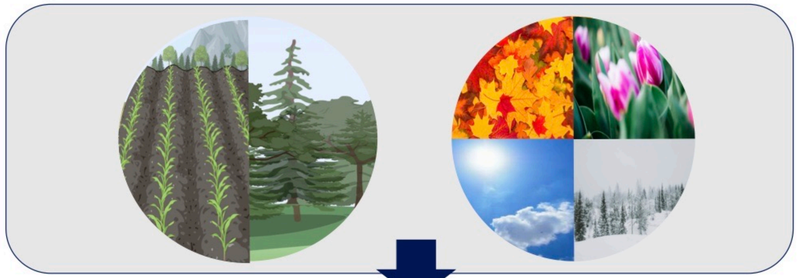
ATMOSPHERIC DEPOSITION
Airborne microbes settle on panels

BIOFILM FORMATION
Microbes attach and grow

SOLUTION

Characterize panel microbiomes to reveal ***land cover*** and ***season effects***

Land Cover + Season



Solar Panel Microbiome

Taxonomy Analysis
Who is there?

Functional Genes
Which traits?

REAL WORLD IMPACT

Restore

Increase solar efficiency by mitigating biosoiling

Predict

Link land use and climate to atmospheric microbiomes

Inform

Apply outdoor framework to indoor microbiomes